Roll No.

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharma (Sem–3)

PHYSICAL PHARMACEUTICS-I

Subject Code : BP-302T

M.Code : 93324

Date of Examination : 03-06-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Define briefly :**

   a) Mixtures

   b) Raoult's Law

   c) Crystalline powders

   d) Dielectric constant

   e) H.B

   f) Interfacial tension

   g) Solubilization

   h) Protein Binding

   i) Glassy state

   j) Solvation

SECTION-B

2. Classify surfactants. Give factors effecting solubilisation. Give methods for the determination of surface tension.

3. Distinguish between :

   a) Contact angle and Spreading coefficient

   b) CMC and Detergency

   c) Wetting agents and Solubilising agents

4. Write short note on :

   a) Solubility of liquid in liquid

   b) Distribution law and its applications.

SECTION-C

5. Explain the diffusion principles in biological systems.

6. Write short note on critical point and liquid crystals.

7. How dissociation constant and dipole moment determined?

8. Explain adsorption at liquid interface.

9. Classify complexes.

10. What is buffered isotonic solution?

11. Enumerate the thermodynamic treatment of stability constant.

12. Give application of buffers with examples of each.

13. Write short note on :

    a) Surface free energy

    b) Relative humidity

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

July-2023